

Pruning the ARL Sensor Interface: a $6.7\times$ Per-Step Speedup with Bit-Identical Trajectories

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1. The cost

The `social.grid` environment advertises **66,223 sensors and 11,250 actuators**: 9,637 pandapower elements (2,294 buses, 2,595 lines, 2,465 sgens, 1,932 loads, 345 transformers) at roughly seven attributes each. This follows from the grid model and the `powergrid` module, so every experiment in the repository sees it. Agents subscribe to **31**; the remaining 99.95 % are built, serialised, shipped across the message bus and discarded, once per step, for the whole of training.

2. Where the filtering happens



The filter runs *between* enumeration and entity creation: a pruned sensor is never wired, stepped or serialised, so the cost is not reduced but never incurred. Two properties make this safe rather than merely fast. `palaestrAI` validates every agent subscription against the advertised sensors at setup and raises before the first step, so a run that starts has full agent observability *by construction* — the filter can abort a run, never silently starve a learner. And it preserves the survivors’ relative order, so summation order in reward computation, and hence its floating-point result, is unchanged.

3. Validation

Four arms were generated from one source run document by one documented mutation each, then run for four scripted phases $\times$ 60 steps. At every step we compared the SHA-256 of the fire raster, the structural telemetry and all 1,932 load sensors, reporting the number of cells whose state differs. Agreement is exact, not within a tolerance.

The `grid_probe` agent was removed from every arm. Its `DummyMuscle` writes `actuator.space.sample()` into `Powergrid-0.0-load-0-0.p_mw`: a random active-power setpoint each step, from a gymnasium `Box` seeded from OS entropy. With it present, two byte-identical configurations diverge and no comparison means anything.

	Arms differ by	Cells differing
Reproducibility control	nothing (identical configurations)	0
Treatment	the filter, on vs. off	0
Sensitivity control	fire wind speed +3.6 %	360

The sensitivity control perturbs the fire driver’s `base_speed` ($14 \rightarrow 14.5$ m/s) with terrain held identical, and must diverge. It is what gives the two zeros meaning: without it, “identical” reported twice is equally consistent with a comparison that detects nothing. Arms were compared as *stored* run documents rather than configuration files — treatment and baseline differ in exactly ten leaves: two identifiers, and the filter entries in each of the four phases.

4. Result

Running each arm at two lengths (8 and 60 steps) determines total = 4 (startup + steps $\times$ per_step) exactly, separating fixed from marginal cost (Fig. 1). Per-step cost falls from 9.77 s to 1.45 s, a factor of **6.73**; per-phase startup is unchanged ($53.8 \rightarrow 53.4$ s). Noise between two bit-identical runs was 0.6 %, so this is not a scheduling artefact. Because fixed cost is untouched, any end-to-end ratio depends on episode length and understates the effect.

5. Limitations

Scripted policies only: the DRL phases still contain `grid_probe`, so they are not reproducible against themselves and bit-comparison does not apply; equivalence for learners rests on Section 2. The store retains only load sensors (identically in both arms), so equivalence is established on the outcome-relevant projection. One scenario, one seed.

6. Reproducing this

```
python analysis/make_whitelist_configs.py # arms, from one source document
./run_whitelist_validation.sh             # ~2 h: all arms, both controls
python analysis/cost_figure.py            # Figure 1
```

The runner aborts before reporting the treatment if the reproducibility control diverges or the sensitivity control does not, and refuses to start if the arm documents have drifted from the generator. Each arm keeps its own database (`analysis/compare_runs.py`).

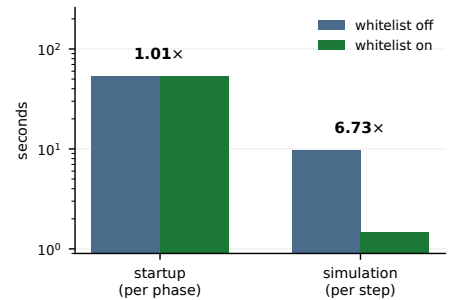


Figure 1: The gain is confined to the per-step term; fixed startup is untouched.